

2.5.9. Product of Maxterms

Each of the 2^n functions of n binary variables can also be expressed as a product of maxterms. To express the Boolean function as a product of maxterms, it must first be brought into a form of OR terms. This may be done by using the distributive law $x+yz = (x+y)(x+z)$. Then any missing variable x in each OR term is ORed with x' .

Example 2-6: Express the Boolean function $F = xy + x'z$ in a product of maxterm form.

$$\begin{aligned}
 F = xy + x'z &= (x+y)(x+z) \quad (\text{Postulate 4(b)}) \\
 &= (x'+x)(z+y) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+x) \cdot (z'+z)) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 4(b)}) \\
 &= ((x'+x) \cdot (x'+z)) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+x) \cdot (x'+z)) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 3(a)}) \\
 &= (1 \cdot (x'+z)) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 5(a)}) \\
 &= (x'+z) \cdot 1 \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 3(b)}) \\
 &= (x'+z) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 2(b)}) \\
 &= (x'+z) \cdot ((z+y) \cdot (z'+z)) \quad (\text{Postulate 2(a)}) \\
 &= ((x'+z) + 0) \cdot (((z+y) + 0) \cdot ((z'+z) + 0)) \quad (\text{Postulate 5(b)}) \\
 &= ((x'+z) + zz') \cdot (((z+y) + yy') \cdot ((z'+z) + x'x')) \quad (\text{Postulate 5(b)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Postulate 4(b)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Postulate 3(b)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Postulate 3(a)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Theorem 4(a)}) \\
 &= ((x'+z) \cdot (x'+y+z')) \cdot (((z+y) \cdot (z'+y')) \cdot ((z'+z) \cdot (z'+y+x'))) \quad (\text{Postulate 3(a)})
 \end{aligned}$$